

High Spatial Resolution Projection Mapping for Visually Consistent Reproduction of Physical Surfaces

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Figure 1: a) Wooden cube with uniform color image projection, b) Plaster cube with wood grain projection at normal pixel density, c) Plaster cube with wood grain projection at high pixel density.

Abstract

In texture representation by projection mapping, it is crucial that the appearance of the presented virtual object matches that of the real object to maintain the quality of the user experience. This study investigates the feasibility of reproducing a virtual object whose appearance perceptually matches that of a real object with fine texture, achieved by using high pixel density projection within an optically reduced projection area of the projector.

CCS Concepts

• Computing methodologies → Mixed / augmented reality.

Keywords

Projection Mapping, Augmented Reality, Perceptual Realism

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1 Introduction

Projection Mapping (PM) enables the manipulation of a real object's appearance, including its color and texture, by overlaying images

from a projector onto an uneven or patterned three-dimensional surface. To ensure the quality of the user experience is maintained, it is essential that the appearance of the virtual object, particularly its texture representation, is accurately aligned with that of the real object.

In replicating the appearance of real objects, it is imperative to depict even the most intricate shapes and textures accurately. Achieving this level of precision in PM has been challenging due to the inadequate spatial resolution of projectors. Prior research has focused on augmenting the perceived resolution of projectors through the superimposition of images from multiple projectors [Okatani et al. 2008]. However, this method of superimposing projection images partially limits the dynamic range of the projector and may diminish its ability to express textures effectively.

In this research, we focus on reducing the projection area by integrating a narrow-angle lens with the projector and increasing the pixel density of the projection image. Additionally, we enable precise texture editing by incorporating techniques such as color compensation. The objective is to present virtual objects that accurately match the appearance of real objects.

Previous research has aimed to replicate the appearance of real objects by reproducing their luminance, depth, color space, depth, and other attributes [Zhong et al. 2021]. Unlike these approaches, which utilize optical see-through displays, this study explores the feasibility of achieving similar results using PM.

2 Proposed System

This study increases the number of pixels per unit area of the projection image to achieve high-definition texture presentation, thereby enhancing pixel density. Additionally, it incorporates essential techniques for texture editing, such as color compensation,

to accurately represent the desired color of the projection image on the projection surface.

2.1 Optical System with Reduced Projection

To increase the pixel density, we design the optics of a projector with a narrow-angle lens to reduce the projection area. When using a standard projector, reducing the distance between the projector and the projection target improves spatial resolution, but no image is formed below the minimum projection distance. To solve this problem, we combined two types of lenses with a narrow-angle lens and a projection lens, thereby enabling high pixel-density projection.

2.2 Color Compensation

To ensure the target color is accurately projected onto the projection surface, we implement a process to compensate for the color projected by the projector by solving the inverse problem of the color transformation process between the color projected by the projector and the color observed by the camera, which acts as a substitute for the observer's vision. This process requires obtaining the reflective properties of the projection surface. The reflection characteristics are determined using a color transformation matrix representing each camera pixel's optical relationship to the corresponding projector pixel [Yoshida et al. 2003].

Once the color transformation matrix is obtained, the projector color that reproduces the target color is given as follows:

$$P_t = F^{-1}(C_t - E) \quad (1)$$

where $C_t \in \mathbb{R}^3$ is the RGB value of the target image, $P_t \in \mathbb{R}^3$ is the RGB value of the projection image reproducing it, $F \in \mathbb{R}^{3 \times 3}$ is the color transformation matrix, and $E \in \mathbb{R}^3$ is the RGB value of the projection surface under ambient light.

In practice, however, cameras that handle only RGB information cannot acquire the complete spectral information of the target object, which may lead to color deviations in the compensation results. Therefore, in addition to the method above, an optimization calculation is performed to match the target image and the captured image more closely, reproducing the desired color with greater accuracy. Let $j \in \mathcal{J} := 1, 2, \dots, m$ be the index of the pixels. The projector input image intensity $x^j \in [0, 1]$ is given as follows:

$$(x^j)^* = \arg \min_{x^j} \sum_{j \in \mathcal{J}} (r^j - Y^j)^2, \text{ s.t. } x^j \in [0, 1]$$

where $r^j \in \mathbb{R}_+$ ($j \in \mathcal{J}$) is the target image and $Y^j \in \mathbb{R}$ ($j \in \mathcal{J}$) is the captured image.

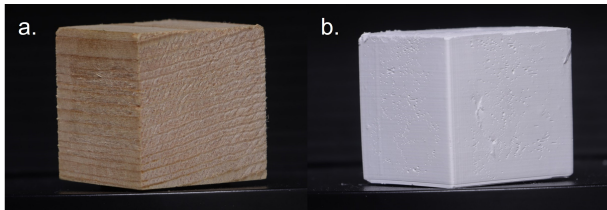


Figure 2: Projection targets: a) Wooden cube, b) Plaster cube

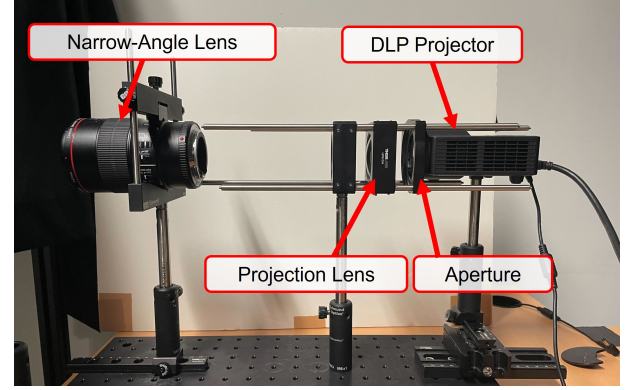


Figure 3: Reduction projection system with narrow-angle lens

This optimization problem is solved by applying a camera-in-the-loop feedback process with the projection results, thereby obtaining a projection image that accurately reproduces the target image.

3 Results and Conclusion

The proposed system uses high pixel density projection to present an indistinguishable virtual object from a real object. In this paper, we compared a real wooden cube (Figure 2 (a)) with a plaster cube (Figure 2 (b)) that reproduces the appearance of a wooden cube by projecting the grain onto it.

Figure 1 presents the result of projecting a uniform color image onto a wooden cube and the color compensation results onto a plaster cube under two different pixel density conditions. One condition using an optical system with reduced projection (Figure 3) involves a high pixel density projection (106 mm diagonal, 359 ppi). The other condition features a normal pixel density (590 mm diagonal, 65 ppi). The compensation result at normal pixel density did not allow for the expression of fine-grain patterns, making the difference from the real wooden cube obvious. On the other hand, although it is the author's subjective evaluation, when the compensation was performed at high pixel density, almost no difference could be perceived in terms of color and pattern fineness compared to the real wooden cube. Therefore, increasing pixel density by optically reducing the projection area is considered effective in presenting a virtual object that is indistinguishable from a real object.

Acknowledgments

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